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VLDUS: Vision-language distilled unseen synthesizer for zero-shot object detection

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ABSTRACT

Generative methods have shown promising performance on zero-shot object detection (ZSD) by synthesizing visual features of unseen classes from semantic embeddings. Although largely compensating for the lack of training samples, they learn the feature synthesizer of unseen classes solely based on limited training data of seen classes, leading to poor diversity and generalization ability of synthesized unseen samples. To overcome this challenge, we develop a Vision-Language Distilled Unseen Synthesizer, namely VLDUS, to build up a novel knowledge distillation-based feature generation paradigm for ZSD. To regulate the synthesized feature space, VLDUS designs two complementary generative distillation strategies that can distill rich image-text knowledge from a pre-trained CLIP model to the synthesizer. To mitigate the over-fitting towards seen classes, VLDUS performs feature-aligned generative distillation on the discriminator's embedding space to methodically learn from the CLIP embedding space, and thus endows the synthesizer with strong generalization ability. To guarantee the intra-class diversity of synthesized unseen features, relation-aligned generative distillation is further performed to distill the diversified image-text correlations from pre-trained CLIP model to the synthesizer. Extensive experiments on MS COCO 2014, PASCAL VOC 2007/2012 and DIOR demonstrate that the proposed VLDUS can generate unseen features of both high intra-class diversity and inter-class separability, and thus outperforms state-of-the-art methods by a large margin on both ZSD and GZSD tasks. Our code is publicly available at <https://github.com/Xcxnh/VLDUS>.

1. Introduction

Zero-shot object detection (ZSD) addresses the challenge of localizing and recognizing objects from novel categories not seen during training (Hayat et al., 2020; Rahman et al., 2018b; Wang et al., 2024a). Traditional object detection methods (Cai & Vasconcelos, 2018; Ren et al., 2015; Xie et al., 2021; Yuan et al., 2024) rely on extensive labeled training data to learn representations of object classes and their visual characteristics, which is impractical for the vast number of object categories in real-world scenarios. ZSD overcomes this limitation by leveraging auxiliary information (e.g., textual descriptions, attributes, semantic embeddings) to bridge the gap between seen and unseen classes. By exploiting the shared embedding space, ZSD models can generalize knowledge to localize and classify objects from novel categories. This enables the detection of previously unseen objects, making ZSD valuable for real-world

applications with a growing variety of objects. While ZSD primarily focuses on detecting objects from unseen classes, the detection system often encounters a mix of both seen and unseen classes simultaneously in practical scenarios. This more challenging setting, known as Generalized ZSD (GZSD) (Liu et al., 2022; Rahman et al., 2019), pushes the boundaries of traditional ZSD methods by requiring the model to detect objects from both seen and unseen classes.

In recent years, ZSD/GZSD has received significant research interest from the computer vision community. Prior studies have explored two distinct approaches, i.e., mapping-based and generative-based methods. Mapping-based methods, initially popular in ZSD, aim to establish a mapping between the visual feature space and the semantic embedding space (Bansal et al., 2018; Rahman et al., 2020; Yan et al., 2020). By mapping unseen classes into this shared space, the model can generalize its knowledge to detect and recognize objects from novel categories.

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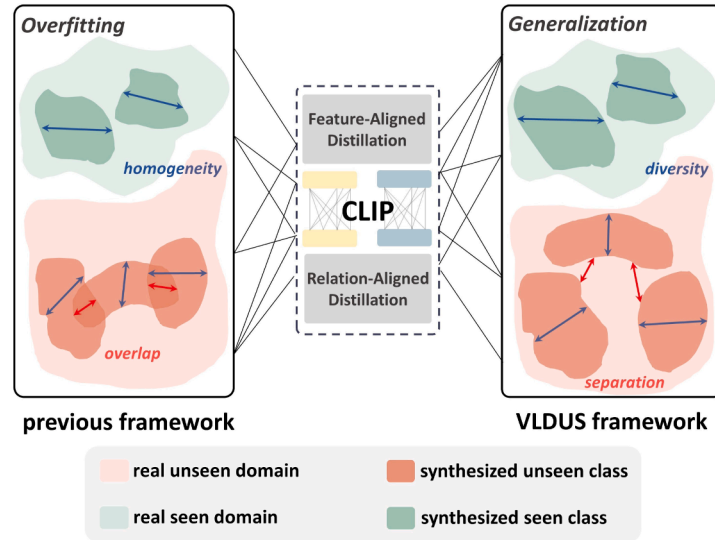


Fig. 1. Comparison of the visual feature distribution between previous framework and the proposed VLDUS. Left part: Previous generative methods for ZSD train the synthesizer with limited data from seen classes, leading to limited diversity and generalization ability of synthesized features. Middle and right part: For this issue, our VLDUS performs vision-language distillation from the pre-trained CLIP model on both feature-level and relation-level to regulate the visual feature space.

However, due to the lack of unseen-class data, the learned mapping function could suffer from severe bias towards seen categories, leading to degraded ZSD/GZSD performance. For this issue, researchers have begun to explore generative-based methods for ZSD/GZSD (Hayat et al., 2020; Huang et al., 2022; Zhao et al., 2020; Zhu et al., 2020), which synthesize samples of unseen classes by leveraging generative models. Both the generated unseen samples and the real seen samples contribute to the training of classifier, thereby alleviating the bias problem towards seen classes.

Compared to the conventional zero-shot image recognition, the zero-shot object detection task faces two unique challenges: 1) *intra-class variation*. Objects of the same category exhibit great variation in appearance and size. 2) *inter-class co-occurrence*. An image usually contains multiple objects, where these objects usually come from different categories or even span the seen and unseen domains. This requires the generated features to have sufficient intra-class diversity, as well as strong cross-domain generalization ability. However, the feature synthesizers of existing generative ZSD methods struggle to meet these two requirements. This is because they learn the underlying distribution of unseen-class data solely based on the visual-semantic relationship of seen classes, leading to low intra-class diversity and insufficient generalization ability as illustrated in the left part of Fig. 1. On one hand, for each seen class, only a limited number of samples are available for training the generative model, resulting in intra-class homogeneous sample generation. On the other hand, due to the domain gap between seen and unseen classes (Rahman et al., 2019), the learned data distribution of the synthesizer is still prone to over-fitting to seen classes, leading to limited generalization ability. As a result, the unseen features generated by previous ZSD synthesizers inevitably deviate from the ground truth, thereby degrading the performance of trained unseen classifier. Thus, it's far from optimal to directly transfer the feature synthesizer trained on seen classes to unseen classes.

The human brain memorizes rich vision-language associations, and uses relevant knowledge in past memories to imagine and identify new things (Schacter et al., 2012). Based on this insight, an intuitive idea is to optimize the synthesized unseen feature space by resorting to the rich knowledge within pre-trained vision-language models (Du et al., 2022; Zhang et al., 2024). Recent advances show that the Contrastive Language-Image Pre-training (CLIP) model (Radford et al., 2021), pre-trained on 400 million image-text pairs, yields a large vision-language embedding space with impressive transferring capabilities (Patashnik et al., 2021; Rao et al., 2022). Inspired by this, we propose VLDUS, a

novel vision-language knowledge distillation-based feature generation framework, to regulate the ZSD synthesizer by distilling knowledge from the pre-trained CLIP feature space, as shown in the middle and right part of Fig. 1. To pursue diverse and generalizable feature generation, we develop a vision-language distilled unseen synthesizer for VLDUS, equipped with two innovative generative distillation strategies. First, it performs feature-aligned generative distillation to improve the synthesizer's generalization ability to unseen classes. Instead of directly aligning the generated visual feature space, we turn to regulate the feature space of our discriminator by mimicking that of the CLIP model, and simultaneously conduct real-fake cross distillation to challenge the discriminator's learning process. Second, in addition to feature-level distillation, relation-aligned generative distillation is further performed to distill the rich image-text correlation knowledge from CLIP to our synthesizer. It improves the diversity of synthesized samples by learning from the diversified correlation knowledge within the pre-trained model.

The contributions of the present work can be summarized as follows:

- We reveal the key challenge in generative-based ZSD, *i.e.*, the synthesizer's over-fitting towards seen classes, and establish a vision-language knowledge distillation based feature generation framework to pursue diverse and generalizable feature synthesis.
- We develop a novel vision-language distilled unseen synthesizer for ZSD, namely VLDUS, which regulates the synthesized visual feature space by distilling the rich image-text knowledge from pre-trained CLIP model.
- Two complementary generative distillation strategies are designed to overcome the aforementioned over-fitting challenge, which leverage the CLIP embedding space at both feature-level and relation-level to guide the training of our generative model.
- Comprehensive experiments on four different dataset splits for ZSD/GZSD, *i.e.*, MS COCO 65/15, MS COCO 48/17, PASCAL VOC and DIOR, demonstrate the superior performance of the proposed approach.

2. Related works

2.1. Generative zero-shot learning

In recent years, the field of zero-shot learning has seen a surge of methods for visual feature synthesis (Li et al., 2021; Wang et al., 2024b),

driven by the impressive sample synthesis capabilities of deep generative networks. These methods, known as generative zero-shot learning approaches, synthesize visual samples for unseen classes based on semantic class information, effectively transforming the zero-shot recognition task into a traditional supervised learning task and circumventing the impact of domain shift (Ye et al., 2021) on the model. Xian et al. (2018) employed Wasserstein GAN (Arjovsky et al., 2017) as the baseline model and introduced an additional classifier and classification loss function to enhance the inter-class discriminability of the synthesized unseen class samples. This is one of the earliest works that use generative adversarial networks to solve the zero-shot learning problem. Subsequently, Felix et al. (2018) proposed a cycle-consistent based Wasserstein GAN (cycle-WGAN), mapping the synthesized fake samples back to the semantic space for the reconstruction of the original semantic features, and introducing a multi-modal cycle consistency loss term to enhance the semantic consistency of the synthesized visual samples. Li et al. (2019a), in response to the issue of feature confusion between seen and unseen classes, proposed an Alleviating Feature Confusion GAN (AFC-GAN) model, which introduced a boundary loss to improve the discriminability between seen and unseen classes. More recently, Chen et al. (2023) developed a dynamic semantic prototype evolving method to align the empirically predefined semantic prototypes and the real prototypes. Hou et al. (2024) proposed a visual-augmented dynamic semantic prototype method to boost the generator to learn accurate semantic-visual mapping by fully exploiting the visual-augmented knowledge into semantic conditions. In this paper, we also employ a generative model to synthesize visual samples of classes not seen during training, yet our goal is to address the more challenging task of generative-based zero-shot object detection.

2.2. Zero-shot object detection

The zero-shot object detection (ZSD) task was first introduced in Bansal et al. (2018), Rahman et al. (2018b), and has received tremendous research interest in recent years. Most existing works adopt mapping-based strategies to approach the ZSD/GZSD task. These methods focus on learning an optimal projection function from visual to semantic space (Rahman et al., 2018b; Yan et al., 2020), from semantic to visual space (Zhang et al., 2020a,b), or learning a common space between visual space and semantic space over seen categories (Gupta et al., 2020; Li et al., 2019b). This mapping allows the model to generalize to unseen classes based on the learned associations. However, due to the lack of unseen visual training samples, these mapping-based methods can inevitably suffer from severe bias problem to seen categories. More recently, generative-based approaches were proposed to address the bias problem. For example, GTNet (Zhao et al., 2020) first presented a synthesizer-assisted paradigm to balance the classifier training with synthesized unseen-class features. SU (Hayat et al., 2020) improved the framework by simplifying the synthesizer and instead using loss components to manipulate the synthesis. RRFS (Huang et al., 2022) added more components to reinforce intra-class diversity and inter-class separability. Wang et al. (2024a) revealed the scale-varying challenge in ZSD and proposed a scale-aware unseen imagineer to lead the way of a scale-aware ZSD paradigm. Despite their empirical success, the feature synthesizer of these methods can easily overfit and bias to limited training data of seen classes, which gives rise to our VLDUS, i.e., a vision-language knowledge distillation based feature generation framework for ZSD.

2.3. Knowledge distillation for object detection

Since the introduction of Knowledge Distillation (KD) by Hinton et al. (2015), it has been widely used to pass dark knowledge from complicated teachers to compact students, enabling students to maintain strong performance as teachers (Gou et al., 2021). In addition to the classification task (Liu et al., 2024; Shang et al., 2023), several methods

have been proposed that use KD for object detectors. The previous literature on KD-based object detection exhibits great diversity, such as soft label distillation, feature distillation, distillation network and strategy design. For example, some related works guide student model training using soft labels or pseudo-label output of teacher models, so as to improve the performance of student model and reduce their dependence on ground truth labels (Guo et al., 2020; Wang et al., 2021). Another KD strategy that can also effectively improve the performance of object detection models is feature distillation in intermediate feature layers (He et al., 2021; Qi et al., 2021). In general, these methods use the features output from the middle layer of teacher model to supervise the training of student model. Besides, researchers also explored other strategies for further improvement, including various teacher-student network structures (Xing et al., 2022), multiple teacher models (Kuang & Liu, 2021) and self-distillation (He et al., 2022). Despite their efficacy, these methods focus on facilitating models developed for general object detection tasks that adopt the pure object detection network architecture. However, when it comes to the ZSD task equipped with generative models, the KD strategies developed by previous object detection methods become infeasible. Thus, in this work, we aim to explore effective generative distillation strategies for the GAN synthesizer that can enhance its unseen-class feature synthesis ability in ZSD.

3. The proposed methodology

3.1. Problem definition

We start by the problem definition of ZSD and GZSD task. The object class space is composed of a seen label set $\mathcal{Y}^s = \{y_i^s\}_{i=1}^{c_s}$ of c_s categories and an unseen label set $\mathcal{Y}^u = \{y_i^u\}_{i=1}^{c_u}$ of c_u categories, where $\mathcal{Y}^s$ and $\mathcal{Y}^u$ follow a disjoint constraint, i.e., $\mathcal{Y}^s \cap \mathcal{Y}^u = \emptyset$. Each class is represented as a d_s -dimensional semantic embedding $\mathbf{w}$, denoted as $\mathbf{W}^s = [\mathbf{w}_1^s, \mathbf{w}_2^s, \dots, \mathbf{w}_{c_s}^s]$ for seen categories and $\mathbf{W}^u = [\mathbf{w}_1^u, \mathbf{w}_2^u, \dots, \mathbf{w}_{c_u}^u]$ for unseen categories. In ZSD/GZSD setting, the visual examples corresponding to the unseen categories $\mathcal{Y}^u$ are unavailable during training; thus, the training dataset consists only of samples from the seen categories. The training dataset can be denoted as $D_{tr} = \{I_i^tr\}_{i=1}^{N_{tr}}$ with a total number of N_{tr} images, where I_i^tr refers to the i -th image in D_{tr} . Each image contains several seen-class objects with the corresponding category labels and bounding box coordinates. The testing set is denoted as $D_{te} = \{I_i^{te}\}_{i=1}^{N_{te}}$ with a total number of N_{te} images. During ZSD testing, we only need to detect objects from unseen categories; while for the more challenging GZSD evaluation, both seen-class and unseen-class objects are expected to be detected.

3.2. Framework overview

To approach the ZSD/GZSD problem, we develop a vision-language distilled feature generation framework, i.e., VLDUS. As illustrated in Fig. 2, the VLDUS framework is mainly composed of a basic detection model for seen object prediction and a conditional generative model for unseen feature synthesis. The overall training pipeline of VLDUS can be summarized into the following three stages.

Train Detector with Seen-Class Annotations. Given an arbitrary image I as input, we first employ bottom convolutional layers to generate intermediate convolutional activations as feature maps $\mathbf{F}$. Subsequently, Region Proposal Network (RPN) is employed to generate a set of candidate object proposals with varying sizes. Next, RoI-pooling is applied to extract the visual features of varying-sized proposals from $\mathbf{F}$, which are then projected to fixed-dimensional representations. Finally, the region features are passed through two modules, i.e., object classifier and bounding box regressor, for class label and box coordinate prediction respectively. Once trained, it can be employed to extract seen-class objects and the corresponding visual features from the training dataset.

Train Synthesizer with Seen-Class Features. Given the extracted region features, class labels and the corresponding semantic embeddings

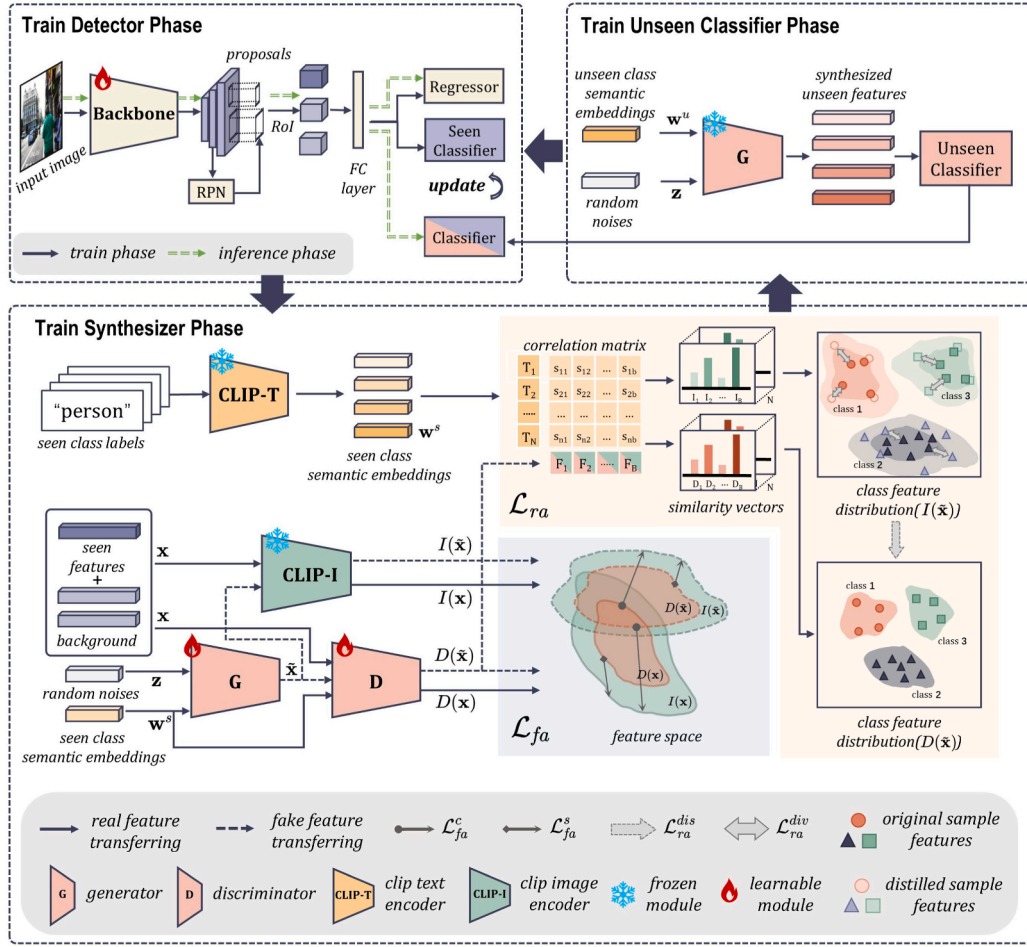


Fig. 2. Illustration of the proposed VLDUS framework. The training pipeline mainly includes three stages: train detector with seen-class annotations, train synthesizer with seen-class features, train unseen classifier with synthesized features. The synthesizer is optimized with both GAN training objectives and two complementary generative distillation components, i.e., FAGD and RAGD highlighted with blue and yellow background, respectively.

from seen classes, we aim to build a unified generative model that can synthesize high-quality features of unseen-class objects, thereby transforming ZSD into the conventional detection problem. To achieve this goal, we develop a Vision-Language Distilled Unseen Synthesizer to learn the mapping relationship between visual and semantic domain, under the guidance of the pre-trained CLIP model. Once trained, its generator network can produce an arbitrary number of unseen object features conditioned on the semantic embedding of each unseen class.

Train Unseen Classifier with Synthesized Features. With the synthesized unseen features and their corresponding class labels, we can update the parameters of the classifier trained in the first stage. The classifier can be endowed with the ability to distinguish unseen-class objects, thus forming a new ZSD detector.

3.3. Vision-language distilled unseen synthesizer

The key to generative ZSD is learning a conditional generative model that can synthesize unseen-class features with the corresponding semantic embeddings. In this part, we elaborate the formulation of our vision-language distilled unseen synthesizer for generalizable and diverse feature synthesis.

Overview. The synthesizer employs Wasserstein GAN as the basic component (Arjovsky et al., 2017), and is composed of two neural networks that interact with each other, i.e., the generator and discriminator. Specifically, the generator can be formulated as $G: \mathcal{W} \times \mathcal{Z} \rightarrow \mathcal{X}$, where $\mathcal{Z}$ and $\mathcal{W}$ refer to Gaussian noise and semantic embedding space respec-

tively; $\mathcal{X}$ denotes the generated feature space. Given random Gaussian noise $\mathbf{z} \sim \mathcal{N}(0, 1) \in \mathcal{Z}$ and semantic embedding $\mathbf{w}_y$ of category $y \in \mathcal{Y}^s$ as input, the generator G learns to generate the CNN features $\tilde{\mathbf{x}} \in \mathcal{X}^s$ of class y . The discriminator $D: (\mathcal{X} \cup \tilde{\mathcal{X}}) \times \mathcal{W} \rightarrow [0, 1] \in \mathbb{R}$ transforms both the real and synthesized features, along with the corresponding class semantic embeddings, into a real value in order to distinguish between them. The synthesizer is trained in an adversarial manner, where the generator and discriminator compete against each other. To pursue generalizable and diverse feature synthesis, we optimize the synthesizer with vanilla WGAN training objectives, classification loss and two complementary generative distillation components.

Training Objectives. During training, the generator tries to fool the discriminator by generating samples that are mistakable for real, while the discriminator tries to accurately distinguish between real and fake ones. In this way, they are alternately trained via the following min-max optimization problem:

$$\min_{\theta_g} \max_{\theta_d} \mathcal{L}_{WGAN}(\theta_g, \theta_d) + \alpha_1 \mathcal{L}_C(\theta_g, \theta_c^*) + \alpha_2 \mathcal{L}_{fa}(\theta_g^F, \theta_d, \Theta_I^*) + \alpha_3 \mathcal{L}_{ra}(\theta_g^F, \theta_d, \Theta_I^*). \quad (1)$$

The training objective in Eq. (1) summarizes the full optimization in a compact form, which can be further decomposed into two explicit and separate update steps for the generator and discriminator as follows:

$$\text{Update } D: \max_{\theta_d} \left[\mathcal{L}_{WGAN}(\theta_g^F, \theta_d) + \alpha_2 \mathcal{L}_{fa}(\theta_g^F, \theta_d, \Theta_I^*) + \alpha_3 \mathcal{L}_{ra}(\theta_g^F, \theta_d, \Theta_I^*) \right], \quad (2)$$

$$\text{Update } G : \min_{\theta_g, \theta_d} [\mathcal{L}_{WGAN}(\theta_g, \theta_d^F) + \alpha_1 \mathcal{L}_C(\theta_g, \theta_d^*)], \quad (3)$$

where θ_g and θ_d refer to the training parameters of G and D respectively; θ_c^* represents the optimal parameters of the pre-trained classifier C over seen categories; Θ_I^* refers to the pre-trained weights of the CLIP image encoder. The superscripts F and $*$ indicate that the corresponding parameters are frozen during loss computation. $\mathcal{L}_{WGAN}$ denotes the loss for Wasserstein GAN, while $\mathcal{L}_C$ is the classification loss that enhances the discriminant ability of the synthesized features. $\mathcal{L}_{fa}$ and $\mathcal{L}_{ra}$ denote the generative distillation loss for feature and relation alignment respectively. The hyperparameters α_1 , α_2 and α_3 are employed to balance different loss components.

Let $\tilde{\mathbf{x}}^s$ denote the fake seen features synthesized by $G(\mathbf{w}_y^s, \mathbf{z}|\theta_g)$, the Wasserstein GAN loss can be formulated as:

$$\begin{aligned} \mathcal{L}_{WGAN} = & \mathbb{E}[D(\mathbf{x}^s, \mathbf{w}_y^s|\theta_d)] - \mathbb{E}[D(\tilde{\mathbf{x}}^s, \mathbf{w}_y^s|\theta_d)] \\ & - \lambda \mathbb{E}[(\|\nabla_{\tilde{\mathbf{x}}^s} D(\tilde{\mathbf{x}}^s, \mathbf{w}_y^s|\theta_d)\|_2 - 1)^2], \end{aligned} \quad (4)$$

where $\tilde{\mathbf{x}}^s = \eta \mathbf{x}^s + (1 - \eta) \tilde{\mathbf{x}}^s$ is a momentum term with η sampled from the uniform distribution $\eta \sim U(0, 1)$; λ is the trade-off hyperparameter. The classifier pre-trained on seen categories can be thought of as a regularizer enforcing the generator to construct discriminative features. Thus, we use negative log likelihood to formulate the classification loss as

$$\mathcal{L}_C = -\mathbb{E}[\log p(y|G(\mathbf{w}_y^s, \mathbf{z}|\theta_g); \theta_c^*)], \quad (5)$$

where $p(y|G(\mathbf{w}_y^s, \mathbf{z}|\theta_g); \theta_c^*)$ is computed by the softmax classifier parameterized by θ_c^* . It indicates the probability of the synthesized feature $G(\mathbf{w}_y^s, \mathbf{z}|\theta_g)$ being predicted as its true class label y .

3.4. Generative vision-language distillation

For generative ZSD, generalizability and diversity are two key properties that reflect the quality of feature generation, thus largely determine the ZSD performance. However, existing methods fail to synthesize unseen-class features with sufficient generalizability and diversity due to the two inherent problems: 1) the significant gap between seen-class and unseen-class domain, and 2) the limited seen-class samples for synthesizer training. To this end, we design two generative vision-language distillation strategies to methodically enhance our synthesizer by distilling rich knowledge from the pre-trained CLIP model.

Feature-Aligned Generative Distillation. To address the problem that the synthesized features lack generalization ability to unseen classes, we design Feature-Aligned Generative Distillation (FAGD) module to regulate the discriminator's feature space by critically mimicking the CLIP embedding space.

Given real visual feature $\mathbf{x}^s$ associated with a seen class y^s , we feed it to the CLIP image encoder I , yielding *real-CLIP* feature $I(\mathcal{R}(\mathbf{x}^s)|\Theta_I^*)$, and to the discriminator D , yielding *real-GAN* feature $D(\mathbf{x}^s|\theta_d)$, where $\mathcal{R}(\cdot)$ is the resize operation that transforms the feature dimension from $\mathbb{R}^{1024}$ to $\mathbb{R}^{32 \times 32}$. Given fake visual feature $\tilde{\mathbf{x}}^s = G(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F)$ synthesized for a seen class y^s , we feed it to the CLIP image encoder I , yielding *fake-CLIP* feature $I(\mathcal{R}(\tilde{\mathbf{x}}^s)|\Theta_I^*)$, and to the discriminator D , yielding *fake-GAN* feature $D(\tilde{\mathbf{x}}^s|\theta_d)$. Thus, we can obtain a set of real and fake features, i.e., $\mathcal{X} = \{I(\mathcal{R}(\mathbf{x}^s)|\Theta_I^*), D(\mathbf{x}^s|\theta_d), I(\mathcal{R}(\tilde{\mathbf{x}}^s)|\Theta_I^*), D(\tilde{\mathbf{x}}^s|\theta_d)\}$. The overall loss function of FAGD module can be derived as:

$$\mathcal{L}_{fa} = -\mathcal{L}_{fa}^s(\mathbf{x}^s, \mathbf{w}_y^s, \mathbf{z}|\theta_g^F, \theta_d, \Theta_I^*) - \mathcal{L}_{fa}^c(\mathbf{x}^s, \mathbf{w}_y^s, \mathbf{z}|\theta_g^F, \theta_d, \Theta_I^*), \quad (6)$$

where $\mathcal{L}_{fa}^s$ and $\mathcal{L}_{fa}^c$ are the loss components for same-feature alignment and cross-feature alignment respectively.

First, we align the feature space of CLIP and our GAN discriminator with the same input, which is achieved by calculating the pairwise ℓ_1 loss of *real-GAN*, *real-CLIP* and *fake-GAN*, *fake-CLIP*:

$$\begin{aligned} \mathcal{L}_{fa}^s(\mathbf{x}^s, \mathbf{w}_y^s, \mathbf{z}|\theta_g^F, \theta_d, \Theta_I^*) = & |D(\mathbf{x}^s|\theta_d) - I(\mathcal{R}(\mathbf{x}^s)|\Theta_I^*)| \\ & + |D(G(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F)|\theta_d) - I(\mathcal{R}(G(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F))|\Theta_I^*)|. \end{aligned} \quad (7)$$

The minimization of $\mathcal{L}_{fa}^s$ enables generalizable knowledge distillation from the pre-trained CLIP model, thereby enhancing the synthesizer's generalization ability to unseen classes.

Second, considering the over-fitting problem caused by limited training data, we further conduct real-fake cross-alignment over different input features in $\mathcal{X}$, i.e., *real-GAN*, *fake-CLIP* and *fake-GAN*, *real-CLIP*, to challenge the discriminator's learning process, i.e.,

$$\begin{aligned} \mathcal{L}_{fa}^c(\mathbf{x}^s, \mathbf{w}_y^s, \mathbf{z}|\theta_g^F, \theta_d, \Theta_I^*) = & |D(\mathbf{x}^s|\theta_d) - I(\mathcal{R}(G(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F))|\Theta_I^*)| \\ & + |D(G(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F)|\theta_d) - I(\mathcal{R}(\mathbf{x}^s)|\Theta_I^*)|. \end{aligned} \quad (8)$$

In Eq. (8), $\mathcal{L}_{fa}^c$ crosses the real and fake features for GAN learning. This yields a more challenging task for the discriminator to distinguish real and fake samples, thus further mitigates the over-fitting problem.

Relation-Aligned Generative Distillation. To enhance the intra-class diversity of synthesized features, we further perform Relation-Aligned Generative Distillation (RAGD) during the synthesizer training process, which aims to distill the diversified image-text correlations from the pre-trained CLIP model.

Given synthesized visual features $\tilde{\mathbf{x}}^s = G(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F)$, we first encode them with the image encoder of CLIP to get $I(\mathcal{R}(G(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F))|\Theta_I^*)$, and then compute the pairwise similarity between the encoded visual features and the class semantic embeddings $\mathbf{W}'$ to get a correlation matrix, denoted as $\mathbf{S}_t$. Note that the classes in $\mathbf{W}'$ are composed of both a background class and all the foreground seen classes. Similarly, we compute the pairwise similarity between the discriminator encoded visual features $D(G(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F)|\theta_d)$ and the class semantic embeddings $\mathbf{W}'$, denoted as $\mathbf{S}_s$. The two correlation matrices are computed as follows:

$$\mathbf{S}_t = \frac{I(\mathcal{R}(G(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F))|\Theta_I^*)\mathbf{W}'^T}{\|I(\mathcal{R}(G(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F))|\Theta_I^*)\mathbf{W}'^T\|_2}, \quad (9)$$

$$\mathbf{S}_s = \frac{D(G(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F)|\theta_d)\mathbf{W}'^T}{\|D(G(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F)|\theta_d)\mathbf{W}'^T\|_2}, \quad (10)$$

where $\mathbf{W}'^T$ is the transpose of $\mathbf{W}'$ and $\|\cdot\|_2$ refers to ℓ_2 -normalization.

To pursue diversified feature generation, the correlations between images and texts encoded by CLIP should be as diverse as possible. Thus, we first introduce the pairwise diversity loss $\mathcal{L}_{ra}^{div}$ to encourage the feature diversity in $\mathbf{S}_t$, which is designed to minimize the pairwise similarity of visual-semantic correlation vectors encoded by CLIP. Notably, the visual features utilized in this component are the fake features generated by G . Thus, $\mathcal{L}_{ra}^{div}$ can encourage G to produce non-repetitive content, which can be formulated as:

$$\mathcal{L}_{ra}^{div} = \sum_{i=1}^B \sum_{j=1, j \neq i}^B \cos(\mathbf{s}_t^i, \mathbf{s}_t^j), \quad (11)$$

where $\mathbf{s}_t^i$ and $\mathbf{s}_t^j$ correspond to the i -th and j -th row of $\mathbf{S}_t$ respectively; B is the total number of similarity vectors. The function $\cos(\cdot)$ computes the cosine similarity of two vectors.

Then, we establish the knowledge transfer from $\mathbf{S}_t$ to $\mathbf{S}_s$, enabling D to acquire the inherent vision-language correlation encoded in the pre-trained CLIP model. This can be achieved by minimizing the ℓ_1 -based relation distillation loss $\mathcal{L}_{ra}^{dis}$, i.e.,

$$\mathcal{L}_{ra}^{dis} = |\mathbf{S}_t - \mathbf{S}_s|. \quad (12)$$

Notably, the pairwise diversity loss $\mathcal{L}_{ra}^{div}$ in Eq. (11) also benefits the relation distillation process of $\mathcal{L}_{ra}^{dis}$ by transferring more diverse correlations to $\mathbf{S}_s$. Thus, the overall training loss for the RAGD module can be formulated as:

$$\mathcal{L}_{ra} = -\mathcal{L}_{ra}^{dis}(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F, \theta_d, \Theta_I^*) - \beta \mathcal{L}_{ra}^{div}(\mathbf{w}_y^s, \mathbf{z}|\theta_g^F, \Theta_I^*), \quad (13)$$

where $\mathcal{L}_{ra}^{div}$ and $\mathcal{L}_{ra}^{dis}$ in Eq. (13) refer to the pairwise diversity loss and relation distillation loss respectively. The hyperparameter β is employed to balance the two loss components.

3.5. Overall training strategy

The proposed VLDUS framework is mainly accomplished by three training stages. The first stage is to train the Faster R-CNN detection network for object classification and bounding box regression, which is optimized by the following objective:

$$\mathcal{L}_{s_1} = \mathcal{L}_{cls} + \mathcal{L}_{reg}, \quad (14)$$

where $\mathcal{L}_{cls}$ and $\mathcal{L}_{reg}$ are the cross-entropy classification loss and smooth ℓ_1 bounding box regression loss respectively. The second stage aims to train the GAN-based synthesizer by the following objective:

$$\mathcal{L}_{s_2} = \mathcal{L}_{WGAN} + \alpha_1 \mathcal{L}_C + \alpha_2 \mathcal{L}_{fa} + \alpha_3 \mathcal{L}_{ra}. \quad (15)$$

After training with Eq. (15), we feed the unseen class embeddings and random noise into the generator so that it can be generalized to unseen classes. For each unseen class, the generator can synthesize an arbitrary number of visual features to solve the data scarcity problem. In the third stage, the synthesized features can be exploited to optimize the standard softmax classifier for unseen classes parameterized by θ_u . The classifier is trained by minimizing the negative log likelihood loss to derive the optimal parameters θ_u^* , which are subsequently employed to update the classifier in the detector, thereby endowing it with the capability to detect unseen classes.

4. Experiments

4.1. Experimental settings

Datasets. We evaluate our method on three well-established benchmark datasets for ZSD, i.e., MS COCO 2014 (Lin et al., 2014), PASCAL VOC 2007/2012 (Everingham et al., 2010), and DIOR (Li et al., 2020). For each dataset, we adhere to the standard seen/unseen class splits established in prior ZSD/GZSD literature (Bansal et al., 2018; Hayat et al., 2020; Huang et al., 2022; Li et al., 2023; Wang et al., 2024a; Zhou et al., 2024) to ensure fair and direct comparability. For MS COCO 2014, which includes a total of 80 classes, we adopt both the 65/15 and 48/17 seen/unseen class splits (Bansal et al., 2018). For PASCAL VOC 2007 and 2012, we adopt a 16/4 seen/unseen split for the 20 classes (Bansal et al., 2018). We train our model using images from both the 2007 and 2012 training sets. For evaluation, we use the 2007 validation and test sets (Hayat et al., 2020). DIOR is a collection of remote-sensing images with 20 classes, where we use a 16/4 seen/unseen split (Huang et al., 2022). **Evaluation Protocols.** For PASCAL VOC and DIOR, we employ mean average precision (mAP) as the evaluation metric with IoU=0.5. For MS COCO, we follow the evaluation protocols in Bansal et al. (2018), Rahman et al. (2018a) to report both recall@100 and mAP with IoU=0.5. To evaluate our method's class-wise performance, we also report the average precision (AP) of each category. For the GZSD setting involving both seen and unseen classes, we report seen-class score, unseen-class score and the Harmonic Mean (HM) (Han et al., 2021) of them to indicate the overall GZSD performance. Notably, HM scores of the mAP and recall@100 metric are denoted as mAP-HM and recall@100-HM respectively.

Implementation Details. We employ Faster R-CNN (Ren et al., 2015) as the basic detection model with ResNet-101 (He et al., 2016) as backbone. For MS COCO/PASCAL VOC/DIOR dataset, we train the Faster R-CNN model, synthesizer, and unseen classifier for 12/18/14, 100/150/150, and 15/25/25 epochs respectively. Moreover, for unseen classifier training, we employ G to synthesize 300/500/500 features for each unseen class on MS COCO/PASCAL VOC/DIOR dataset. The hyperparameter α_1 in Eq. (1) is set to 0.01/0.1/0.1. Besides, α_2, α_3 are set to 0.005 and 0.01, while β in Eq. (13) is set to 0.1. We employ CLIP text encoder (Radford et al., 2021) to obtain the semantic embeddings of all the classes. The generator network G consists of two fully-connected layers, while the discriminator network D is implemented as three fully-connected layers. In order to align with the embedding dimension of the

CLIP image encoder, the last hidden layer of D is set to 512 output dimension. For fair comparison, other settings are consistent with previous works (Hayat et al., 2020; Huang et al., 2022).

4.2. Comparison with state-of-the-arts

Performance on MS COCO. As detailed in Table 1, we present a comprehensive comparison between VLDUS and current top-performing methods over the two splits of MS COCO, where the performance is reported under both ZSD and GZSD settings. For the 48/17 split in the ZSD setting, VLDUS outperforms all competitors by a large margin in terms of both recall@100 and mAP. Specifically, compared to the second-best SeeDS method (Zhou et al., 2023), VLDUS raises the ZSD recall@100 from 55.3% to 66.3% and mAP from 14.0% to 17.7%. For the 65/15 split, our method also shows significant performance gains over all the comparison methods. Compared with the prior state-of-the-art method SAUI (Wang et al., 2024a), our VLDUS improves recall@100 from 77.8% to 78.6% and mAP from 22.1% to 26.3% respectively. These significant improvements demonstrate VLDUS's superior ability to detect objects of previously unseen classes.

The comparison results in GZSD setting have also been presented in Table 1. In terms of the seen-class scores, the proposed VLDUS achieves 63.7% recall@100 and 30.5% mAP for the 48/17 split, 72.2% recall@100 and 38.6% mAP for the 65/15 split, which rank among the best in the table, with only a few scores slightly trailing the top performer. This is reasonable because our VLDUS aims to enhance the generalization ability of the synthesizer to unseen classes, while mitigating the overfitting problem towards seen classes. Although the performance on seen classes may suffer some degradation, VLDUS achieves almost the best performance on unseen classes among all the competitors for both the 48/17 and 65/15 split. Compared with the second-best method, it gains a relative improvement of 19% mAP for the 48/17 split, and 1% recall@100 and 23% mAP for the 65/15 split. Thus, VLDUS significantly alleviates the performance gap between seen and unseen classes to achieve more balanced detection results. This has also been confirmed by VLDUS's superior performance on the HM metric for GZSD. For the 48/17 split, the recall@100-HM of VLDUS reaches the second-best at 61.8%, only 0.1% lower than the best at 61.9%, while the mAP-HM is increased from 21.6% to 22.1% by our method. For the 65/15 split, our method raises recall@100-HM from 70.3% to 71.4% and mAP-HM from 27.0% to 31.2%, indicating a significant leap over the prior state-of-the-art SAUI (Wang et al., 2024a).

We additionally present the class-wise AP performance for unseen classes in ZSD setting for MS COCO 65/15 split. As shown in Table 2, VLDUS not only exhibits the highest mAP in the ZSD setting but also achieves the best and second-best AP in 7 and 3 out of 15 unseen classes. For instance, in terms of categories such as "airplane", "parking meter", and "toilet", our method elevates the AP from the second-best scores of 27.8% to 45.4%, 6.5% to 14.7%, and 18.7% to 23.2% respectively. These remarkable improvements highlight VLDUS's advantage in generating high-quality visual features for the vast majority of unseen classes. **Performance on PASCAL VOC.** In Table 3, we compare VLDUS and current leading methods on PASCAL VOC dataset in both ZSD and GZSD settings. In ZSD setting, our method shows a clear advantage on the sofa category, improving from 58.0% to 66.8%, while the other results are slightly inferior to the best-performing method ZSFDet (Zhou et al., 2024). This is because the small size and limited number of categories in the dataset, which restricts our method's ability to fully exploit the enhancement of intra-class and inter-class relationships. Nonetheless, our method achieves the best result of 49.9% mAP-HM in the GZSD setting. This is attributed to the fact that our approach effectively mitigates overfitting on seen classes and better balances the performance between seen and unseen classes, leading to an overall improvement.

Performance on DIOR. In Table 4, we conduct a comparative analysis between our approach and current top-performing methods on DIOR. Specifically, our model achieves 12.3% mAP in ZSD setting, and 8.3%

Table 1

Comparison of recall@100 (%) and mAP (%) in ZSD and GZSD settings at IoU = 0.5 across two seen/unseen splits on MS COCO dataset. The best and second-best results are marked bold and underlined respectively. The symbol “*” denotes the prior state-of-the-art method.

Method	Split	ZSD		GZSD					
		recall@100	mAP	recall@100			mAP		
				Seen	Unseen	HM	Seen	Unseen	HM
BLC (Zheng et al., 2020)	48/17	48.8	10.6	57.6	46.4	51.4	42.1	4.5	8.2
PL (Rahman et al., 2020)		43.5	10.1	38.2	26.3	31.2	35.9	4.1	7.4
ConZSD (Yan et al., 2022)		52.4	12.5	<u>65.7</u>	52.4	58.3	45.1	6.3	11.1
RRFS (Huang et al., 2022)		53.5	13.4	59.7	58.8	59.2	42.3	13.4	20.4
TCB (Li et al., 2023)		52.4	11.4	72.0	52.4	60.7	47.3	4.9	9.8
SeedS (Zhou et al., 2023)		55.3	<u>14.0</u>	60.1	60.8	60.5	42.5	<u>14.5</u>	<u>21.6</u>
ZSFDet (Zhou et al., 2024)		54.7	<u>14.0</u>	60.1	60.7	60.4	<u>45.2</u>	14.3	21.4
SAUI* (Wang et al., 2024a)		<u>66.0</u>	13.7	64.7	59.3	61.9	43.3	13.6	20.7
Ours		66.3	17.7	63.7	60.0	<u>61.8</u>	30.5	17.3	22.1
BLC (Zheng et al., 2020)	65/15	54.7	14.7	57.7	53.9	55.8	36.0	13.1	19.2
PL (Rahman et al., 2020)		37.7	12.4	36.4	37.2	36.8	34.1	12.4	18.2
SU (Hayat et al., 2020)		54.0	19.0	57.7	53.9	55.8	36.9	19.0	25.1
RSC (Sarma et al., 2022)		65.1	20.1	58.6	64.0	61.2	37.4	20.1	26.2
ConZSD (Yan et al., 2022)		59.5	18.6	62.9	58.6	60.7	40.2	16.5	23.4
RRFS (Huang et al., 2022)		62.3	19.8	58.6	61.8	60.2	37.4	19.8	26.0
TCB (Li et al., 2023)		59.9	13.8	69.4	59.9	64.3	<u>39.9</u>	18.3	20.5
SeedS (Zhou et al., 2023)		64.0	20.6	59.3	62.8	61.0	37.5	20.9	26.8
ZSFDet (Zhou et al., 2024)		64.2	20.3	59.3	63.1	61.1	37.5	20.5	26.5
SAUI* (Wang et al., 2024a)		<u>77.8</u>	<u>22.1</u>	<u>70.7</u>	<u>69.9</u>	<u>70.3</u>	36.8	<u>21.3</u>	<u>27.0</u>
Ours		78.6	26.3	72.2	70.7	71.4	38.6	26.1	31.2

Table 2

Comparison of mAP (%) and class-wise AP (%) at IoU = 0.5 for generation-based methods on unseen classes of MS COCO 65/15 in ZSD setting. The best and second-best results are marked bold and underlined respectively.

Method	mAP	airplane	train	parking meter	cat	bear	suitcase	frisbee	snowboard	fork	sandwich	hot dog	toilet	mouse	toaster	hair drier
SU (Hayat et al., 2020)	19.0	10.1	48.7	1.2	64.0	<u>64.1</u>	12.2	0.7	28	<u>16.4</u>	19.4	0.1	<u>18.7</u>	1.2	0.5	0.2
PL (Rahman et al., 2020)	12.4	20.0	48.2	0.6	28.3	13.8	12.4	21.8	15.1	8.9	8.5	0.9	5.7	0.0	1.7	0.0
RSC (Sarma et al., 2022)	20.1	22.9	53.3	0.6	64.9	54.3	13.2	1.2	31.2	15.7	<u>22.6</u>	0.0	17.5	2.7	0.7	0.2
RRFS (Huang et al., 2022)	19.8	20.8	<u>53.0</u>	1.3	<u>64.3</u>	55.5	11.6	0.4	<u>31.3</u>	18.0	20.3	0.1	15.2	4.2	0.5	0.6
SAUI* (Wang et al., 2024a)	<u>22.1</u>	27.8	42.4	<u>6.5</u>	53.8	53.2	15.4	<u>31.3</u>	28.9	8.3	24.3	12.4	14.3	10.2	<u>1.2</u>	1.1
Ours	26.3	45.4	46.4	14.7	61.6	67.1	18.7	34.6	34.9	13.8	21.3	<u>7.2</u>	23.2	<u>4.7</u>	0.6	<u>0.9</u>

Table 3

Comparison of mAP (%) at IoU = 0.5 on PASCAL VOC dataset, including class-wise AP (%) in ZSD setting.

Method	ZSD				GZSD			
	car	dog	sofa	train	mAP	Seen	Unseen	HM
SAN (Rahman et al., 2018b)	56.2	85.3	62.6	26.4	57.6	48.0	37.0	41.8
HRE (Demirel et al., 2018)	55.0	82.0	55.0	26.0	54.5	62.4	25.5	36.2
PL (Rahman et al., 2020)	63.7	87.2	53.2	44.1	62.1	-	-	-
BLC (Zheng et al., 2020)	43.7	86.0	60.8	30.1	55.2	58.2	22.9	32.9
SU (Hayat et al., 2020)	59.6	92.7	62.3	45.2	64.9	-	-	-
RRFS (Huang et al., 2022)	60.1	93.0	59.7	49.1	65.5	47.1	49.1	48.1
SeedS (Zhou et al., 2023)	60.4	95.3	65.9	53.8	68.9	48.5	50.6	49.5
SAUI (Wang et al., 2020)	45.7	93.5	62.1	53.7	65.5	47.2	48.9	48.0
ZSFDet* (Zhou et al., 2024)	63.7	94.3	58.0	60.8	69.2	48.5	50.8	49.6
Ours	55.3	93.4	66.8	49.5	66.3	54.2	46.3	49.9

Table 4

Comparison of mAP (%) at IoU = 0.5 in ZSD and GZSD settings on DIOR dataset.

Method	ZSD	GZSD		
		Seen	Unseen	HM
PL (Rahman et al., 2020)	0.4	4.3	0.0	0.0
BLC (Zheng et al., 2020)	1.1	6.1	0.4	0.8
SU (Hayat et al., 2020)	10.5	30.9	2.9	5.3
RRFS* (Huang et al., 2022)	11.3	30.9	3.4	6.1
Ours	12.3	22.9	8.3	12.2

Table 5

Ablation studies for different modules of VLDUS on DIOR dataset.

Method	ZSD	GZSD		
		Seen	Unseen	HM
VLDUS w/o $\mathcal{L}_{fa}$ & $\mathcal{L}_{ra}$	10.2	22.9	4.8	7.9
VLDUS w/o $\mathcal{L}_{fa}$	11.2	22.9	6.4	10.0
VLDUS w/o $\mathcal{L}_{ra}$	11.4	22.9	7.4	11.1
VLDUS _{full}	12.3	22.9	8.3	12.2

unseen mAP, 12.2% mAP-HM in GZSD setting. Compared to the second-best RRFS (Huang et al., 2022), our model shows a significant advantage in both ZSD and GZSD settings, particularly in mAP-HM where we observe a growth of 100% from 6.1% to 12.2%. Those remarkable scores indicate VLDUS can generalize well from seen to unseen classes, even in remote sensing imagery.

4.3. Ablation study

Quantitative Analysis. In this section, we conduct ablation studies on DIOR dataset to evaluate the effect of each module. The results are reported in Table 5. We exclude the Feature-Aligned Generative Distillation ($\mathcal{L}_{fa}$), the Relation-Aligned Generative Distillation ($\mathcal{L}_{ra}$), and both of them respectively, to form three ablation models. The removal of the

FAGD module (i.e., VLDUS w/o $\mathcal{L}_{fa}$), leads to a decrease in ZSD mAP by 1.1%, GZSD unseen class mAP and mAP-HM by 1.9% and 2.2%. This highlights the FAGD module's role in mitigating overfitting and improving the generalization ability of VLDUS across seen and unseen classes. Without the RAGD module (i.e., VLDUS w/o $\mathcal{L}_{ra}$), the ZSD mAP drops by 0.9%, while the GZSD unseen class mAP and mAP-HM drop by 0.9% and 1.1%. This phenomenon indicates that the RAGD module can effectively enhance the feature diversity, thereby driving high-quality feature generation. The two modules jointly benefit the full VLDUS model (i.e., VLDUS_{full}) with 2.1% absolute increase in ZSD mAP, 3.5% in GZSD unseen class mAP and 4.3% in GZSD mAP-HM, compared to the ablated model VLDUS w/o $\mathcal{L}_{fa}$ & $\mathcal{L}_{ra}$. This proves that the two generative distillation modules cooperate and contribute to the significant performance boost of the proposed VLDUS.

Visualization Analysis. To intuitively show the effect of each component, we visualize the synthesized unseen-class features on the DIOR dataset using t-SNE (Van der Maaten & Hinton, 2008) in Fig. 3. As shown in Fig. 3(a), features synthesized by the base model exhibit poor inter-class separation, where some features lying outside their corresponding clusters. In Fig. 3(b), by integrating the FAGD component to the base model, the synthesized feature distribution demonstrates better clustering pattern with clearer inter-class separation. This benefits from the feature-level distillation strategy of the FAGD module, which can enhance D 's discrimination difficulty and simultaneously drive G to produce higher-quality features. In Fig. 3(c), the integration of RAGD module yields a more dispersed feature distribution of each class, resulting in an increased intra-class diversity of features. We can attribute this to the diversified image-text correlation knowledge distilled from the pre-trained CLIP model in the RAGD module. The full VLDUS model inherits the advantages of both modules to synthesize unseen-class features with superior clustering patterns (i.e., satisfying both intra-class diversity and inter-class separation), as shown in Fig. 3(d). It demonstrates that the

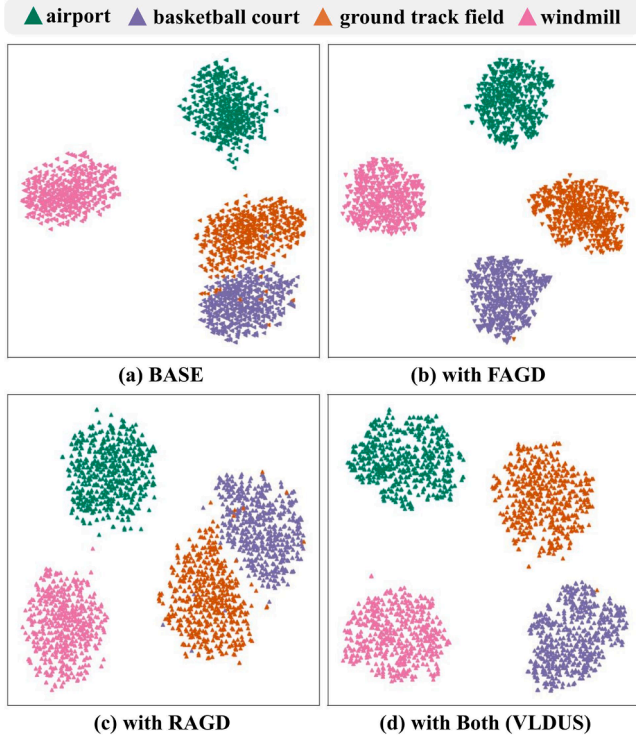


Fig. 3. t-SNE visualization of the synthesized unseen-class features on DIOR dataset: (a) BASE results without two distillation modules (VLDUS w/o $\mathcal{L}_{fa}$ & $\mathcal{L}_{ra}$), (b) results with only the FAGD module (VLDUS w/o $\mathcal{L}_{ra}$), (c) results with only the RAGD module (VLDUS w/o $\mathcal{L}_{fa}$), and (d) results with both FAGD and RAGD modules (VLDUS_{full}).

two key modules of VLDUS work synergistically to improve the synthesis quality of unseen class features.

Hyperparameter Selection. To optimize the model performance, we adopted grid search as the hyperparameter selection strategy. Specifically, based on the conclusions of related works (Hayat et al., 2020; Huang et al., 2024) and the empirical results from our preliminary experiments, we predefined the value ranges of key hyperparameters, thereby reasonably narrowing down the search space. For example, the weight of the classification module α_1 ranges from 0.001 to 0.02 for the MSCOCO dataset and from 0.01 to 0.2 for the VOC/DIOR datasets. The weights of the FAGD module (α_2) and RAGD module (α_3) range from 0.001 to 0.02 across all datasets, and these two parameters necessitate joint tuning. Additionally, the weight of the pairwise diversity loss β ranges from 0.001 to 1 across all datasets. In order to obtain a model that achieves the best balance in performance on both seen and unseen categories, we adopt the mAP-HM under the Generalized ZSD setting as the evaluation metric for hyperparameter selection.

In Fig. 4, we present the results of different hyperparameters choices on the MSCOCO 65/15 dataset because it is the largest dataset with the most diverse categories among the three datasets, thus making its selection results the most representative. As shown in Fig. 4, our method performs best with $\alpha_1 = 0.01$, $\alpha_2 = 0.005$, $\alpha_3 = 0.01$, and $\beta = 0.1$, yielding a mAP-HM of 31.2%. By comparing the results under different hyperparameter settings, we observe that the mAP-HM performance remains relatively stable, with fluctuations within 2.3% across all settings. These results show that our method has a relatively low sensitivity to these hyperparameters and can maintain a quite stable performance under different configurations. This indicates that the modules we proposed are effective when integrated into the model and will not be overly affected by minor adjustments of the hyperparameters.

CLIP Training Strategies. In this section, we investigate whether the parameters of the CLIP model should be frozen during the distillation-based training process. Since CLIP has been pre-trained on large-scale datasets, we considered two different configurations: CLIP parameters are frozen or fine-tuned during distillation. To evaluate the effects of these two settings, we conducted comparative experiments across all datasets. As shown in the results of Table 6, freezing the CLIP model consistently yielded better results. Compared to the fine-tuning setting, where CLIP is jointly trained with the student network, frozen CLIP improved mAP by at least 0.5% and up to 2.2% under the ZSD setting, and mAP-HM by at least 0.5% and up to 1.6% under the GZSD setting. We attribute this to the CLIP model's prior pre-training on large-scale datasets. During the distillation process, the CLIP model acting as the teacher needs to provide strong and stable representation capabilities. Frozen CLIP allows the teacher model to retain the knowledge and patterns learned from large-scale datasets, thereby benefiting the learning process of the student model. In contrast, fine-tuning CLIP may disrupt its original representational power, affecting the effectiveness of knowledge transfer, and consequently limiting the performance of the student model.

Table 6

Comparison of mAP (%) at IoU = 0.5 in ZSD and GZSD settings on all datasets, with Trainable/Frozen CLIP parameter strategies during distillation.

CLIP Model	Dataset	ZSD	GZSD		
			Seen	Unseen	HM
Trainable	MSCOCO 48/17	16.4	30.4	16.1	21.0
	MSCOCO 65/15	24.1	38.5	24.0	29.6
	PASCAL VOC	64.7	53.9	43.7	48.3
	DIOR	11.8	22.9	7.8	11.7
Frozen	MSCOCO 48/17	17.7	30.5	17.3	22.1
	MSCOCO 65/15	26.3	38.6	26.1	31.2
	PASCAL VOC	66.3	54.2	46.3	49.9
	DIOR	12.3	22.9	8.3	12.2

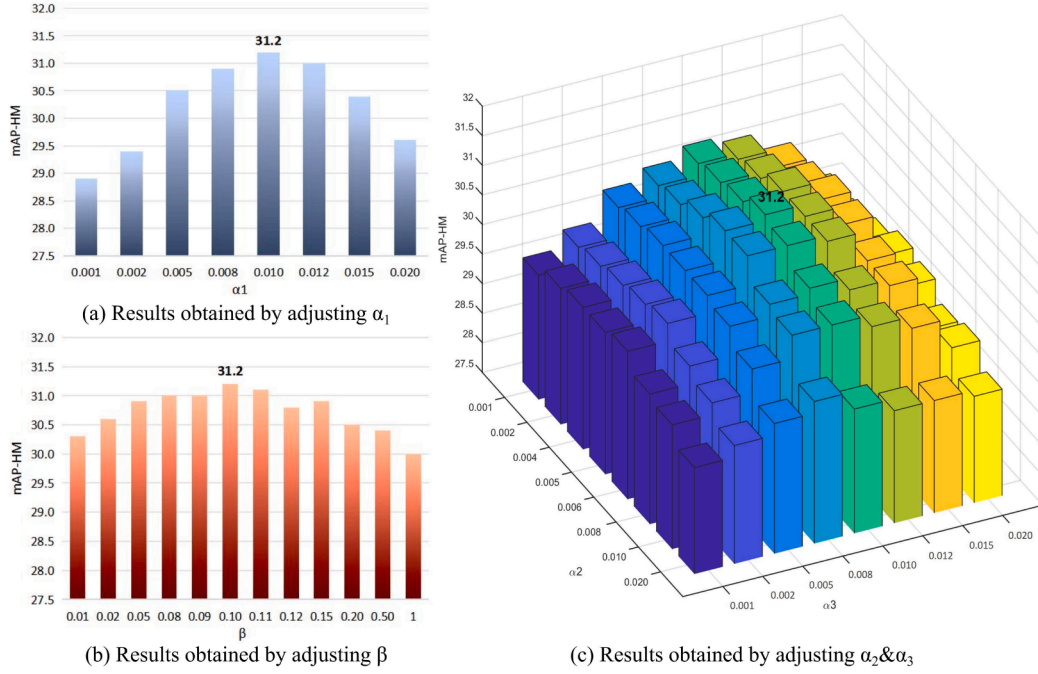


Fig. 4. Results of different hyperparameters choices (α_1 , α_2 & α_3 , β) on the MSCOCO 65/15 dataset.

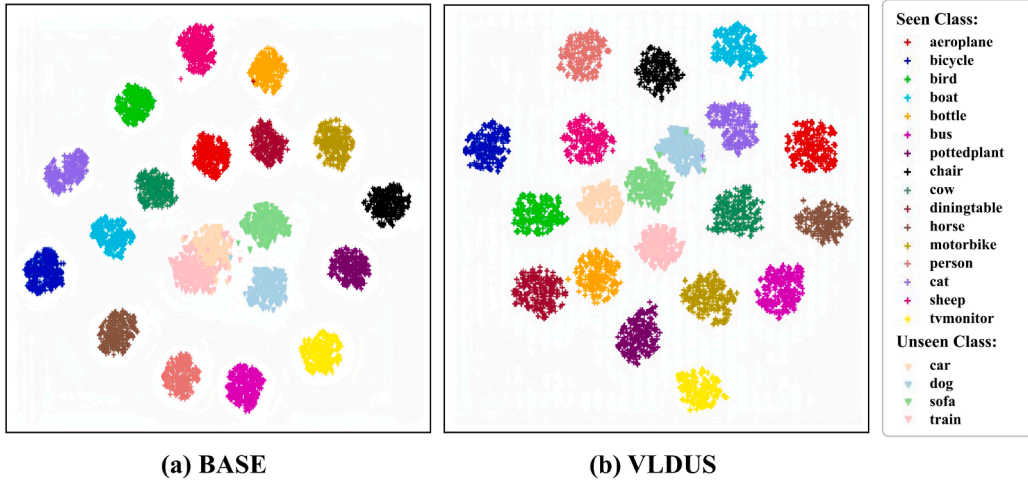


Fig. 5. The t-SNE visualization of the synthesized features for all classes on the PASCAL VOC dataset.

4.4. Qualitative results

Feature Synthesis Visualization. To qualitatively prove our contribution, we visualize the synthesized feature distributions for all classes on the PASCAL VOC dataset. Fig. 5(a) and Fig. 5(b) present the visual feature distribution of our baseline model and the proposed VLDUS respectively. The base result in Fig. 5(a) shows distinct clustering for seen classes but not for unseen ones. For example, the two unseen classes, “car” and “train”, exhibit severe overlap, making them difficult to distinguish. This indicates that the vanilla generator’s feature space is overly biased towards seen classes, leading to poor generalization ability to unseen classes. Moreover, the synthesized features for all categories have shown issues with homogeneity and lack of intra-class variation. With VLDUS, as illustrated in Fig. 5(b), the clustering pattern of the synthesized features significantly improves across all classes, achieving a good trade-off between intra-class diversity and inter-class distinctiveness. This phenomenon indicates that the proposed VLDUS can not only enhance the synthesizer’s generation capacity for unseen

classes but also increase the synthesis diversity, thereby significantly mitigating the overfitting issue in GANs. These enhancements in feature synthesis contribute to the learning of a more discriminative classifier, thereby improving the detection performance for the ZSD/GZSD tasks.

Detection Results. To further demonstrate the effectiveness of our method, we present some detection results from MS COCO 65/15, PASCAL VOC, and DIOR dataset in Fig. 6. For each image, we provide the detection results of both SU (Hayat et al., 2020) and the proposed VLDUS for fair comparison. As shown in Fig. 6, in the ZSD setting, VLDUS can detect the objects of unseen classes precisely while SU misses some of them, particularly on some small objects that are difficult to identify (e.g., “car”, “train”, “ground track field”) and heavily obscured objects (e.g., “toilet”, “suitcase”). Besides, the comparison under the GZSD setting also demonstrates our method’s superior generalization ability to unseen classes, especially on unseen objects that are very similar to seen objects (e.g., “sofa”, “suitcase”). The quantitative results in both ZSD and GZSD setting demonstrate the significant performance of our approach in an intuitive way, establishing a clear gap with SU.

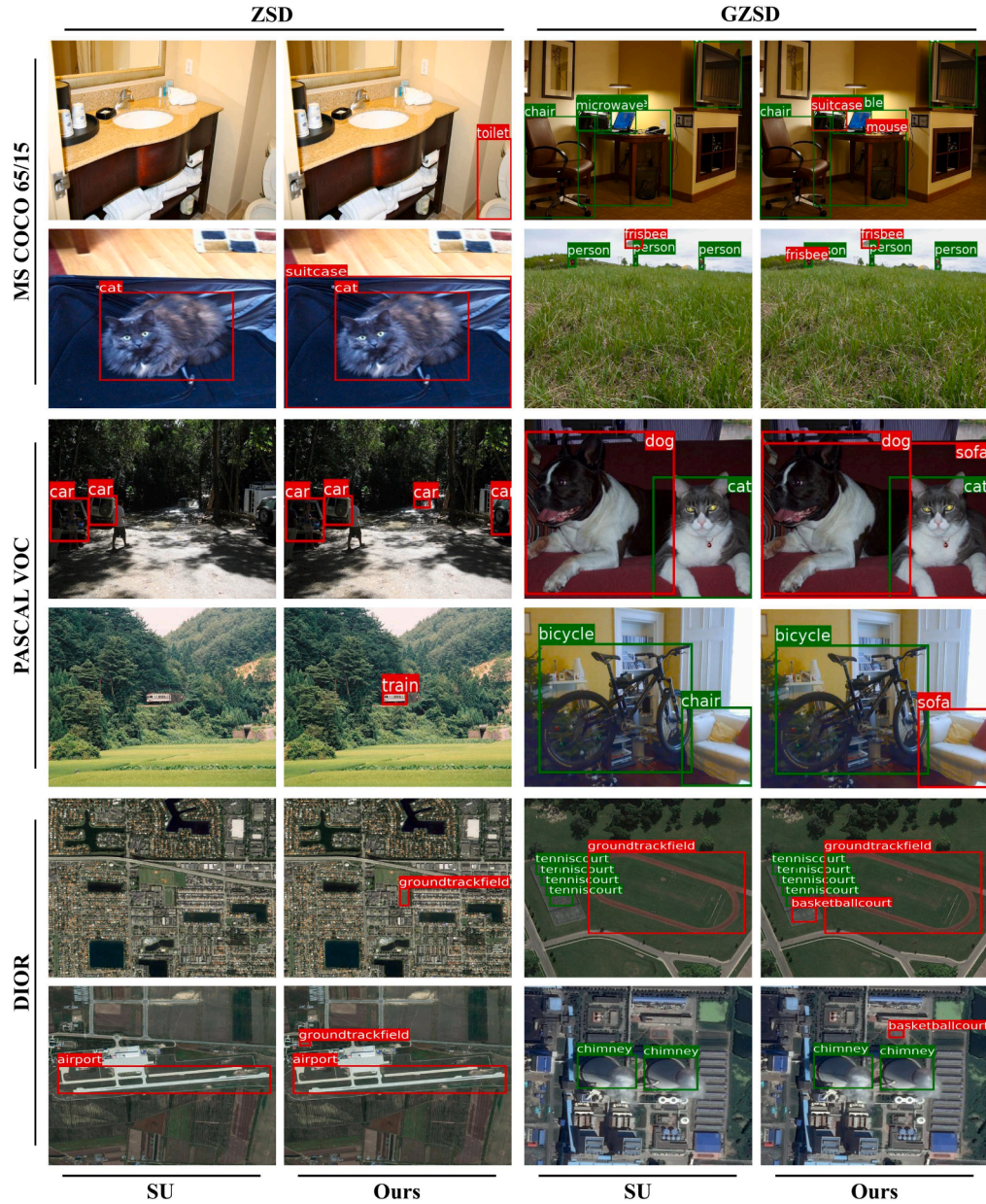


Fig. 6. Qualitative comparison between VLDUS and SU (Hayat et al., 2020) on MS COCO 65/15, PASCAL VOC, and DIOR datasets. The ZSD results and GZSD results are illustrated in the left two columns and the right two columns respectively, with seen-class objects in green boxes and unseen-class objects in red boxes.

5. Conclusion

In this work, we have proposed a novel vision-language distilled unseen synthesizer, namely VLDUS, to lead the way of a vision-language knowledge distillation based feature generation paradigm for ZSD/GZSD. To facilitate diverse and generalizable feature synthesis, VLDUS regulates the synthesized feature space by distilling rich vision-language knowledge from the pre-trained CLIP model. More specifically, it designs two generative vision-language distillation strategies to guide the training of synthesizer by learning from the CLIP embedding space at both feature-level and relation-level. The experimental results demonstrate that the proposed VLDUS significantly surpasses the state-of-the-art methods in both ZSD and GZSD performance. Moreover, the unseen-class features synthesized by our model exhibit a superior trade-off between intra-class diversity and inter-class separation, demonstrating the efficacy of the proposed generative distillation strategies in enhancing the ZSD synthesizer.

CRedit authorship contribution statement

Caixia Yan: Writing – original draft; **Muyan Jiao:** Conceptualization; **Nuohan Xue:** Validation, Resources; **Weizhan Zhang:** Formal analysis; **Jiahao Wang:** Visualization; **Xiaojun Chang:** Project administration; **Feng Tian:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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